

## Catalog Search

Entire Catalog

Search Catalog


☐ Whole Word/Phrase

## Advanced Search

## Catalog Home

### Courses

### Academic Programs

### Schools & Departments

### Dialogues Curriculum

### Dialogues Requirements

### Outline of Dialogues Requirement

## HB 1606: Graduate Outcomes

### General Information

### The Campus

### Admission

### Fees and Financial Aid

### Academic Policies and Procedures

### Truman Policies

### Transferring In and Testing Out

### Student Services

### Student Involvement

### Special Academic Programs

### Honors

### Graduation

## Graduate Studies

### Board of Governors and Administrators

### Emeriti

### Faculty

### Staff

## Former Catalogs

## My Portfolio

## Health Science (BS)


[Return to: Catalog Search](#)

Affiliation: School of Science and Mathematics

## MISSION OF THE HEALTH SCIENCE DEPARTMENT

The mission of the Health and Exercise Sciences Department is to prepare students of strong academic ability and character to become agents of change for the health and well-being of all people.

## PHILOSOPHY OF THE HEALTH SCIENCE DEPARTMENT

The Health Science faculty and staff are dedicated to providing students with a world-class education in a stimulating and supportive environment. This student-centered philosophy is expressed through an emphasis on problem-based learning, practical experience, and scholarly research that promotes a rich learning environment, greatly enhancing student development.

## THE MAJOR

The Bachelor of Science degree in Health Science is enhanced by a broad liberal arts and sciences background in English, mathematics, biological and physical sciences, social sciences, and humanities, together with the specialized courses that comprise the depth of the Health Science major.

The Health Science curriculum is designed to prepare students to promote health in educational or clinical settings in school, worksite, community, or public health environments. The undergraduate degree provides training for entry-level positions in municipal, county, state and national health departments, voluntary health agencies, as well as the curricular background necessary for graduate work in public health, health administration, occupational therapy, medicine, or physician assistant (PA) programs.

One of the unique features of the Truman Health Science undergraduate experience is the opportunity to actively engage in research. A significant number of Health Science majors present their findings at international, national, regional, state, or local professional conferences, or publish their work in peer-reviewed professional journals.

Another unique feature of the Health Science program is its focus on accountability. The health science graduate demonstrates a high level of health knowledge as well as skills based on the Areas of Responsibility of a Certified Health Education Specialist (CHES), as assessed by a senior exit

examination. Health science faculty and the University have selected and approved the CHES examination as the measure of this competency. This feature affords Truman health science students the opportunity to obtain validation of their education through a nationally recognized examination in their specialized field of study. Truman is among only a handful of institutions in the country to provide this opportunity for their health science graduates. Additionally, the Health Science program capstone experience is obtained through the required course of HLTH 440 Program Implementation and Evaluation.

## ADMISSION TO THE HEALTH SCIENCE MAJOR

For students wishing to transfer to the Health Science major, Department Chair approval must be obtained prior to admission.

Students wishing to concentrate in Community, Worksite Public Health must first declare the Pre-Community, Worksite Public Health/Individualized Concentration. Upon completing the below requirements, students can then apply online to concentrate in Community, Worksite Public Health. In order to be in the Community, Worksite Public Health concentration, you must submit an application at <https://www.surveymonkey.com/R/CWPHApp>. The Health Science faculty and Department Chair will approve entrance into the concentration.

### Requirements:

- Overall GPA of 2.5 or higher
- 75 hours completed at the university level
- Completed (**STAT 190 - Basic Statistics** or **STAT 290 - Statistics**) and (**HLTH 255 - Introduction to Community and Public Health** and **HLTH 290 - Public Health Education Principles**) with a grade of “C” or higher.

Suggested deadlines for applying to the concentration are September 20<sup>th</sup> for spring admission and February 20<sup>th</sup> for fall and summer admission. Students who apply after the deadlines will be admitted to the program on a rolling basis.

## OUTCOME STATEMENTS FOR THE HEALTH SCIENCE MAJOR

1. Based on the Core Public Health functions and a strong liberal arts and sciences foundation, demonstrate proficiency in the responsibilities and competencies of the Certified Health Education Specialist (CHES).
  - Assessing needs and capacity for health education/promotion
  - Planning health education/promotion
  - Implementing effective health education/promotion
  - Conducting evaluation and research related to health education/promotion
  - Engaging in advocacy for health education/promotion
  - Effectively communicating health information
  - Coordinating relationships with health partners
  - Behaving in a highly ethical and professional manner
2. Apply CHES competencies through field experiences and community involvement.

# GRADUATION REQUIREMENTS

1. A 2.0 cumulative GPA
2. A 2.0 GPA in all courses completed at Truman
3. A 2.5 major GPA
4. 40 credits of 300- or higher-level courses
5. Minimum 120 total credits of undergraduate coursework
6. Assessment, residency, and all other requirements as outlined in this catalog
7. Grade of "C" or higher must be achieved in core major courses, except **HLTH 400**

## HEALTH SCIENCE CONCENTRATION DESCRIPTIONS

To provide the opportunity for students to explore and develop more specialized interests, the Health Science curriculum requires the selection of one career concentration ranging from 20-25 credits. Courses comprising the program concentrations are based upon the recommendations of faculty members whose expertise is identified with these specialties. Individualized concentrations may also be designed to meet personal goals. The specific courses required for each concentration are listed in the Major Requirements section below.

**Community, Worksite, and Public Health (24 Credits):** The Community, Worksite, and Public Health concentration is accredited by the **Council on Education for Public Health (CEPH)**. The Community Public Health Education Specialist may be employed as a disease prevention/health promotion specialist in voluntary agencies (Red Cross, American Lung Association, Women's Health Clinics, etc.), local agencies (private companies, HMOs, PPOs, hospitals, etc.), county or state agencies (health departments), or federal agencies (CDC, U.S. Department of Health and Human Services, etc.). The specific job may be a one-on-one approach (HIV/AIDS Educator) or it may involve an entire state, as in the case of coalition building. Leadership, ability to work independently, health knowledge, and a vast resource library are the principal ingredients for a Community Public Health Education Specialist. A person in this position would likely be responsible for grant proposals or alternative sources of funding, especially when employed with voluntary agencies. In addition, the health education specialist may have some research and writing responsibilities, thereby necessitating excellent writing and oral communication skills. The expectation of most employers is to minimize or eliminate illness and injury through the assessment, planning, implementation, and evaluation of quality health education/health promotion programs. You must apply to enroll in this concentration. You can apply at: <https://www.surveymonkey.com/R/CWPHApp>.

**Pre-Medical Sciences (20-25 Credits):** Students seeking a career in the medical field need to develop excellent critical thinking skills and complete additional biology, chemistry, and/or physics courses in addition to their Health Science major courses. Thus, the Pre-Medical Sciences Concentration is designed to provide a foundation of science courses for students seeking admission to a variety of health professional programs such as osteopathic medicine, allopathic medicine, physician assistant, dental medicine, optometry, etc. The coursework is designed to prepare the student for the respective graduate medical admission test (Medical College

Admission Test [MCAT], Dental Admission Test [DAT], and Optometry Admission Test [OAT]) students need. Students in this concentration often obtain a minor in biology. To be a viable candidate for admission to a graduate program in health care, a student should maintain a cumulative GPA of 3.25 or higher.

**Pre-Occupational Therapy (20-23 Credits):** This concentration is designed to prepare a student for admission into a graduate program in occupational therapy. Concentration coursework emphasizes courses in psychology. To be a viable candidate for admission to a graduate program in occupational therapy, a student should maintain a cumulative pre-requisite GPA of 3.25 or higher. Truman State University has entered into articulation agreements that provide Truman graduates with priority consideration or guaranteed interviews in a number of occupational therapy programs.

**Pre-Community, Worksite Public Health/Individualized (24 Credits):** The Pre-Community, Worksite Public Health/Individualized Concentration is designed to support future public health professionals and the development of special support areas not available in an established concentration. This concentration must be relevant to the career goals of the student and logically follow from the courses included with the major. This concentration is not a “catch-all” for courses taken that do not fit the published degree requirements. For students in the Individualized Concentration, the Individualized Concentration must be submitted to the Registrar and approved by the academic advisor. These concentrations can include HLTH 460 Internship in Health Education. Examples of individualized concentrations include those for students intending to apply to accelerated nursing programs, music therapy programs, or dietetics programs. Students wanting to concentrate in Community, Worksite, Public Health must apply to that concentration once they meet the requirements specified previously.

## **DEPARTMENTAL HONORS PROGRAM**

### **PURPOSE**

To recognize outstanding students in the major.

### **ELIGIBILITY**

Any Health Science major who attains the following receives Departmental Honors. Students must apply through the Health Science (HS) Department Office during their graduating semester. Applications are available on the Health Science website.

1. Major GPA of 3.5 or higher.
2. Overall GPA of 3.5 or higher.
3. Complete an Health Science-related research project\* and disseminate the results of the research at Truman’s Student Research Conference, in a professional setting, or via a paper that has been submitted or accepted for publication while a Health Science major.
4. Obtain a passing score on the CHES exam.
5. Receive concurrence from health science faculty.

\*The individual was a major contributor to the research project who participated in multiple phases of the project (design, data collection, data analysis, and preparation of the manuscript, poster or presentation). A Health Science faculty member is responsible for verifying the student was a major contributor to the research project.

# DEGREE REQUIREMENTS:

## Dialogues Requirements: 42-61 Credits

---

### Missouri Statute: 0-4 credits

---

## Bachelor of Science Requirement: 7-8 Credits

---

- **BIOL 325 - Human Physiology** Credit(s): 4
- An additional Biology or Psychology course at the 200 level or above or a Chemistry or Statistics course at the 300 level or above.

## MAJOR REQUIREMENTS:

---

The Health Science major consists of two (2) parts: Required Support and Major Requirements. Each student must complete both parts.

### Part I: Required Support: 10 Credits

---

- **BIOL 100 - Biology** Credit(s): 4  
OR **BIOL 107 - Cells, Molecules, and Genes** Credit(s): 4
- **PSYC 166 - General Psychology** Credit(s): 3
- **STAT 190 - Basic Statistics** Credit(s): 3  
OR **STAT 290 - Statistics** Credit(s): 3

### Part II: Major Requirements (Core): 45-50 Credits

---

- **HLTH 255 - Introduction to Community and Public Health** Credit(s): 3
- **HLTH 270 - Health Systems and Consumers** Credit(s): 3
- **HLTH 290 - Public Health Education Principles** Credit(s): 3
- **HLTH 325 - Health Management and Policy** Credit(s): 3
- **HLTH 349 - Research Methods in Health Science** Credit(s): 3
- **HLTH 366 - Program Assessment and Planning** Credit(s): 3
- **HLTH 410 - Health Communication Methods** Credit(s): 3
- **HLTH 440 - Capstone: Program Implementation and Evaluation** Credit(s): 3  
Capstone Experience.
- **HLTH 400 - CHES Exam Review** Credit(s): 1

### Complete one of the following Health Science Concentrations: 20-25 Credits

---

#### Community, Worksite, and Public Health Concentration: 24 Credits

---

Students in this concentration are highly encouraged to take at least one STAT class at the 300-level or above. You must apply

to be in this concentration. You can apply at:

<https://www.surveymonkey.com/R/CWPHApp>.

- **HLTH 405 - Global Public Health** Credit(s): 3
- **HLTH 455 - School Health Programs** Credit(s): 3
- **HLTH 460 - Internship in Health Education** Credit(s): 6
- **HLTH 467 - Introduction to Epidemiology** Credit(s): 3
- **HLTH 480 - Worksite Health** Credit(s): 3

Select at least 2 courses from the following:

- 
- **ES 345 - Exercise Physiology** Credit(s): 3
  - **HLTH 245 - Substance Use Prevention** Credit(s): 3
  - **HLTH 260 - Human Sexuality** Credit(s): 3
  - **HLTH 320 - Patient Education and Clinical Health Promotion** Credit(s): 3
  - **HLTH 361 - Mental Health** Credit(s): 3
  - **HLTH 362 - Environmental Health** Credit(s): 3

Comment:

---

For students who are particularly interested in worksite health promotion, the following courses are recommended in addition to **ES 345 - Exercise Physiology**:

- **ES 445 - Assessment and Prescription**
- **ES 460 - Athletic Conditioning Program Design**

For students who are interested in health administration, a minor in Business Administration is recommended (see **Business Administration Minor**).

### Pre-Medical Sciences Concentration: 20-25 Credits

---

These courses are a foundation for students seeking to enter a clinical graduate or professional program such as Physician Assistant, Dental School, Medical School or Optometry School. However, the student is responsible for researching their desired programs and determining the necessary prerequisites for the program to which they are applying. Those additional courses are not part of the degree or concentration requirements.

- **CHEM 131 - Chemical Principles II** Credit(s): 4
  - **CHEM 320 - Foundations of Organic Chemistry** Credit(s): 4
- OR ((CHEM 329 - Organic Chemistry I** Credit(s): 3) **and** (CHEM 330 - Organic Chemistry I Lab Credit(s): 1 **or** CHEM 333 - Organic Chemistry Lab Credit(s): 2))

Select at least 4 courses from the following:

- 
- **BIOL 104 - Ecology and Evolution of the Organism** Credit(s): 4
  - **BIOL 300 - Genetics** Credit(s): 4
  - **BIOL 304 - General Microbiology** Credit(s): 4
  - **BIOL 321 - Foundations of Genetics** Credit(s): 3
  - **BIOL 330 - Cell Biology** Credit(s): 4

- **BIOL 353 - Pathophysiology** Credit(s): 3
- **CHEM 331 - Organic Chemistry II** Credit(s): 3
- **CHEM 335 - Biochemistry I: Structure and Function** Credit(s): 3
- **HLTH 320 - Patient Education and Clinical Health Promotion** Credit(s): 3
- **HLTH 405 - Global Public Health** Credit(s): 3
- **HLTH 467 - Introduction to Epidemiology** Credit(s): 3
- **PHYS 185 - College Physics I** Credit(s): 4
- **PHYS 186 - College Physics II** Credit(s): 4
- **PSYC 366 - Psychology of Abnormal Behavior** Credit(s): 3
- **PSYC 377 - Developmental Psychology** Credit(s): 3

**Notes:**

---

Students who wish to take the MCAT are recommended to complete a minimum of PHYS 185, PHYS 186, and CHEM 329, CHEM 331, CHEM 333 and CHEM 335.

**Pre-Occupational Therapy Concentration: 20-23 Credits**

---

- **HLTH 200 - Medical Terminology** Credit(s): 2
- **PSYC 366 - Psychology of Abnormal Behavior** Credit(s): 3
- **PSYC 377 - Developmental Psychology** Credit(s): 3
- **SOAN 190 - Introduction to Sociology** Credit(s): 3  
OR **SOAN 191 - Introduction to Anthropology** Credit(s): 3

**Select at least 3 courses from the following:**

---

- **BIOL 365 - Human Anatomy** Credit(s): 3  
AND **BIOL 365L - Human Anatomy Lab** Credit(s): 1
- **CHEM 100 - Chemistry for Contemporary Living** Credit(s): 4  
OR **CHEM 130 - Chemical Principles I** Credit(s): 4
- **ES 282 - First Aid and Injury Care** Credit(s): 3
- **ES 342 - Biomechanics with Lab** Credit(s): 4
- **ES 345 - Exercise Physiology** Credit(s): 3
- **ES 369 - Physical Activity for Individuals with Disability** Credit(s): 3
- **HLTH 320 - Patient Education and Clinical Health Promotion** Credit(s): 3
- **HLTH 361 - Mental Health** Credit(s): 3
- **HLTH 405 - Global Public Health** Credit(s): 3
- **HLTH 467 - Introduction to Epidemiology** Credit(s): 3
- **PHYS 185 - College Physics I** Credit(s): 4
- **PHRE 188 - Ethics** Credit(s): 3  
OR **PHRE 350 - Biomedical Ethics** Credit(s): 3

**Pre-Community Worksite Public Health/Individualized Concentration: Minimum of 24 Credits**

---



This concentration can include

- **HLTH 460 - Internship in Health Education** Credit(s): 6  
Students who do not complete HLTH 460 must include a minimum of five additional credit hours of HLTH courses at the 300-level or above in their concentration. This concentration may include no more than 15 credit hours of courses with the same course prefix (including HLTH).

## Note:

---

Students may not earn both a BS in Health Science and a BS in Exercise Science degree.

## Electives to total minimum of 120 Credits

---

← Return to: [Catalog Search](#)

